DATE	28-02-2026
TEAM ID	LTVIP2026TMID****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

3.2 SOLUTION REQUIREMENTS

The solution requirements for HematoVision: Advanced Blood Cell Classification Using Transfer Learning define the essential functional and non-functional needs of the system to ensure effective performance and real-world applicability in healthcare environments.

1. Functional Requirements:

- The system must accept microscopic blood cell images as input.
- It should preprocess images (resizing, normalization, augmentation) before classification.
- The model must classify different types of blood cells accurately.
- It should use transfer learning with a pre-trained Convolutional Neural Network (CNN).
- The system must display predicted class labels along with confidence scores.
- It should allow evaluation using metrics such as accuracy, precision, recall, and F1-score.
- The system should generate results quickly to support faster diagnosis.

2. Non-Functional Requirements:

- The system must provide high accuracy and reliability.
- It should reduce training time and computational cost.
- The interface must be user-friendly and easy to operate.
- The system should ensure data security and patient confidentiality.
- It must be scalable to handle large datasets in the future.
- The solution should be compatible with standard laboratory systems.

3. Hardware Requirements:

- Computer system with sufficient RAM (minimum 8GB recommended).
- GPU support for faster model training (optional but preferred).
- Storage space for dataset and trained model files.

4. Software Requirements:

- Programming language: Python.
- Deep learning frameworks such as TensorFlow or Keras.
- Image processing libraries like OpenCV.
- Development environment such as Jupyter Notebook or PyCharm.